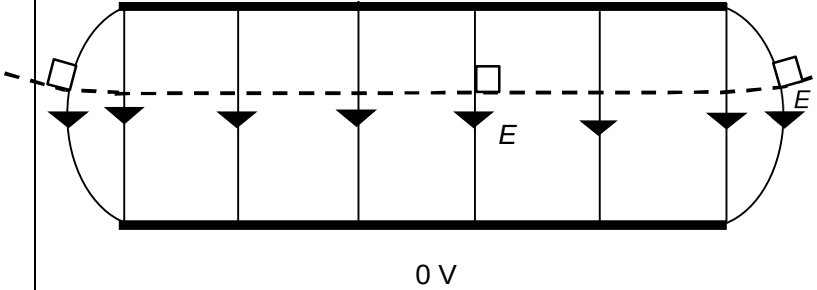
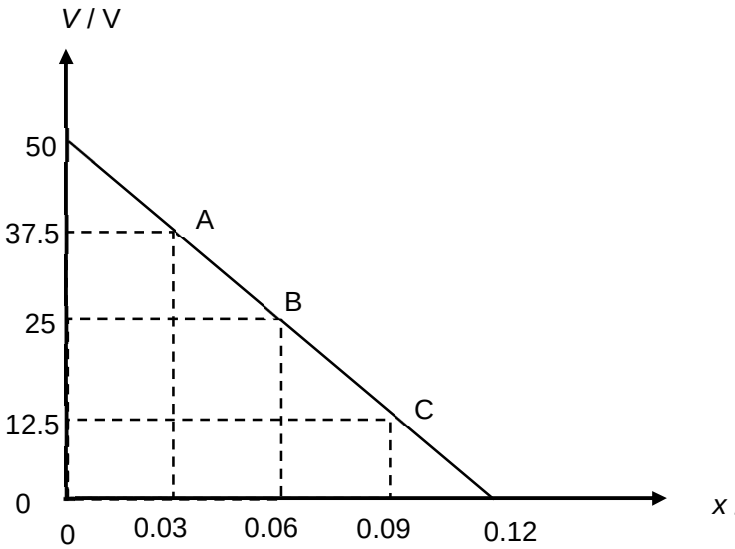


6(a)		[1] equally spaced electric field lines field lines cuts equipotential lines perpendicularly
6(b)		[1] straight line [1] correct values
6(c)	$F = qE = q \frac{V}{d} = (1.60 \times 10^{-19}) \left(\frac{50}{0.12} \right) = 6.7 \times 10^{-17} \text{ N}$	[1] correct electric field E [1] correct answer
6(d)	loss in electric potential energy = gain in K.E. $q(V_A - 0) = \frac{1}{2}mv^2 - \frac{1}{2}m(2 \times 10^4)^2$ $1.60 \times 10^{-19} (37.5 - 0) = \frac{1}{2} (1.67 \times 10^{-27}) (v^2 - (2 \times 10^4)^2)$ $v = 8.7 \times 10^4 \text{ m s}^{-1}$	[1] correct substitution [1] correct answer

$z(a)(i)$	N	[1] sub
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